

Fintech AI – Enterprise System Architecture

SYSTEM OVERVIEW

Fintech AI is an Artificial Intelligence-powered Financial Intelligence and Comparison Platform designed to collect, standardise, compare, analyse, and recommend financial and non-financial products across multiple sectors.

Sectors Covered

- Banking & Insurance
- Telecommunications
- Education & Hospitality
- Utilities & Regulated Prices

Core Capabilities

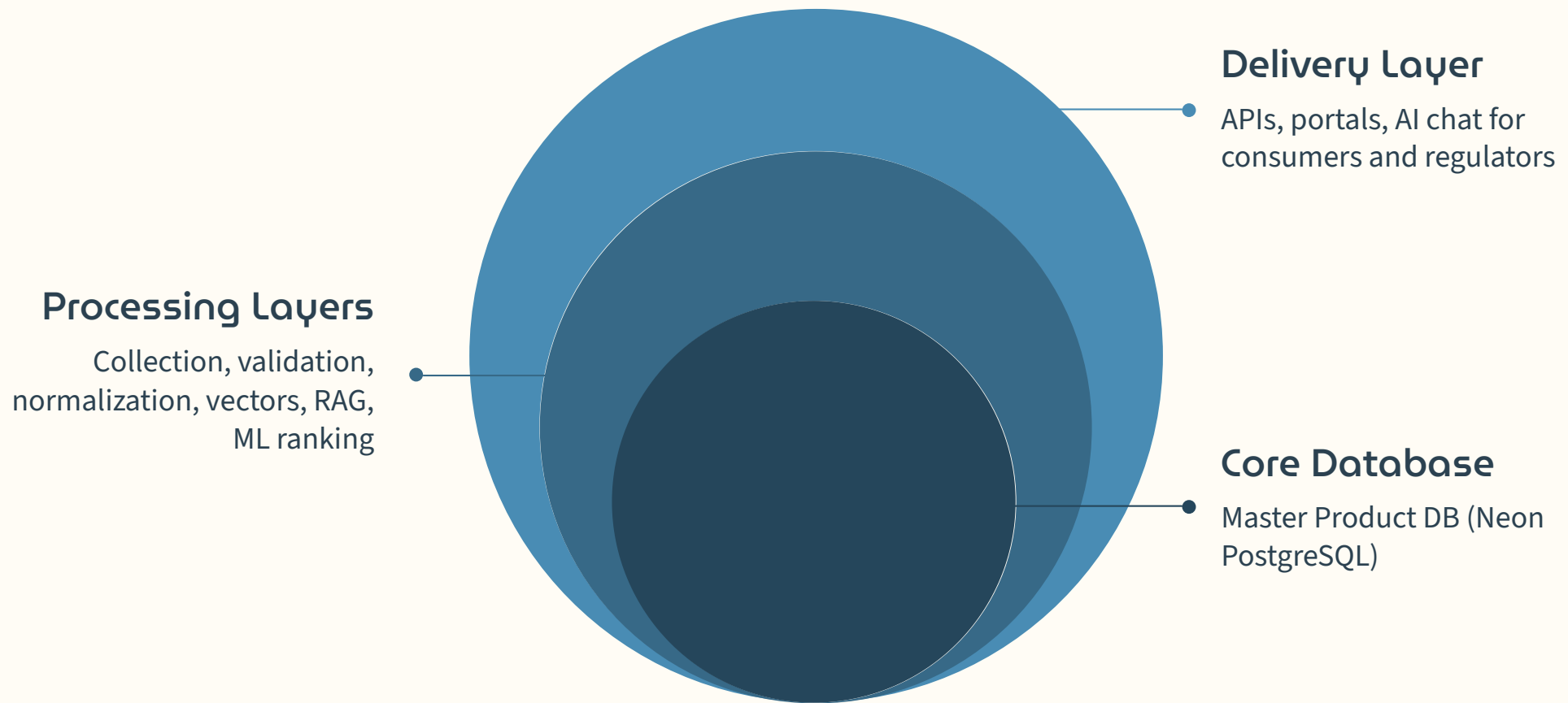
- Real-time web scraping
- AI-powered RAG
- Machine Learning Ranking
- Product Comparison Engines

Interfaces

- Traditional User Interfaces
- Conversational AI Interfaces
- Transparent recommendations
- Data-driven insights

High-Level Architecture

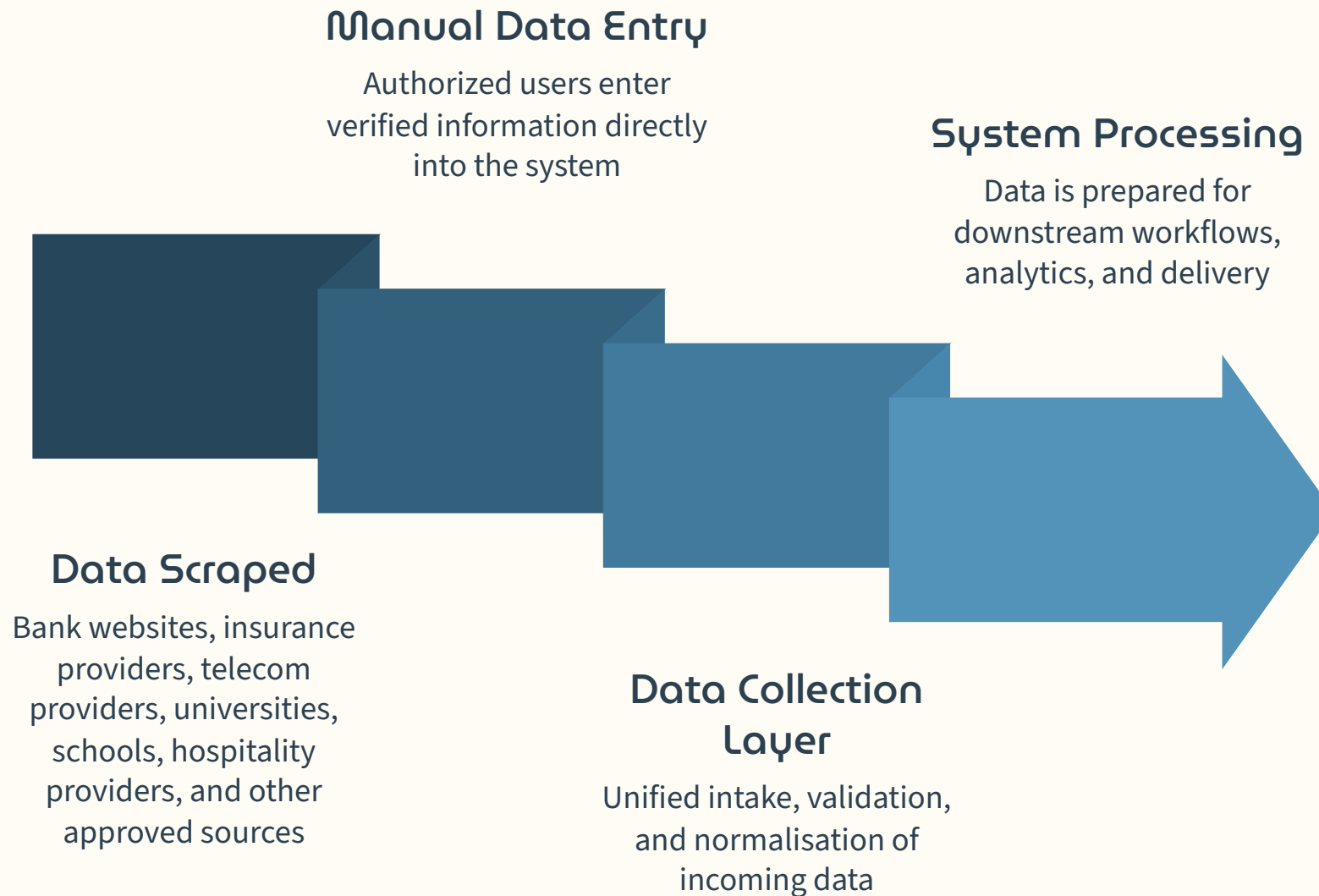
The platform is structured as a layered pipeline — from raw external data sources through to consumer-facing portals — ensuring every recommendation is grounded in validated, normalised, and intelligently ranked information.



Each layer has a distinct responsibility, ensuring separation of concerns, auditability, and scalability across the entire platform.

Data Collection Layer

The system acquires information from two primary sources. The real-time scraping engine continuously collects structured and unstructured data from approved providers across all supported sectors.



Manual Data Management

Authorised users may manually enter or update information within the platform. Manual data is treated as a **trusted source** and is assigned higher confidence scores than scraped data, ensuring data quality and reliability.

1

Developers

Maintain system-level data and platform configurations.

2

Managers

Upload approved market information and validate product records.

3

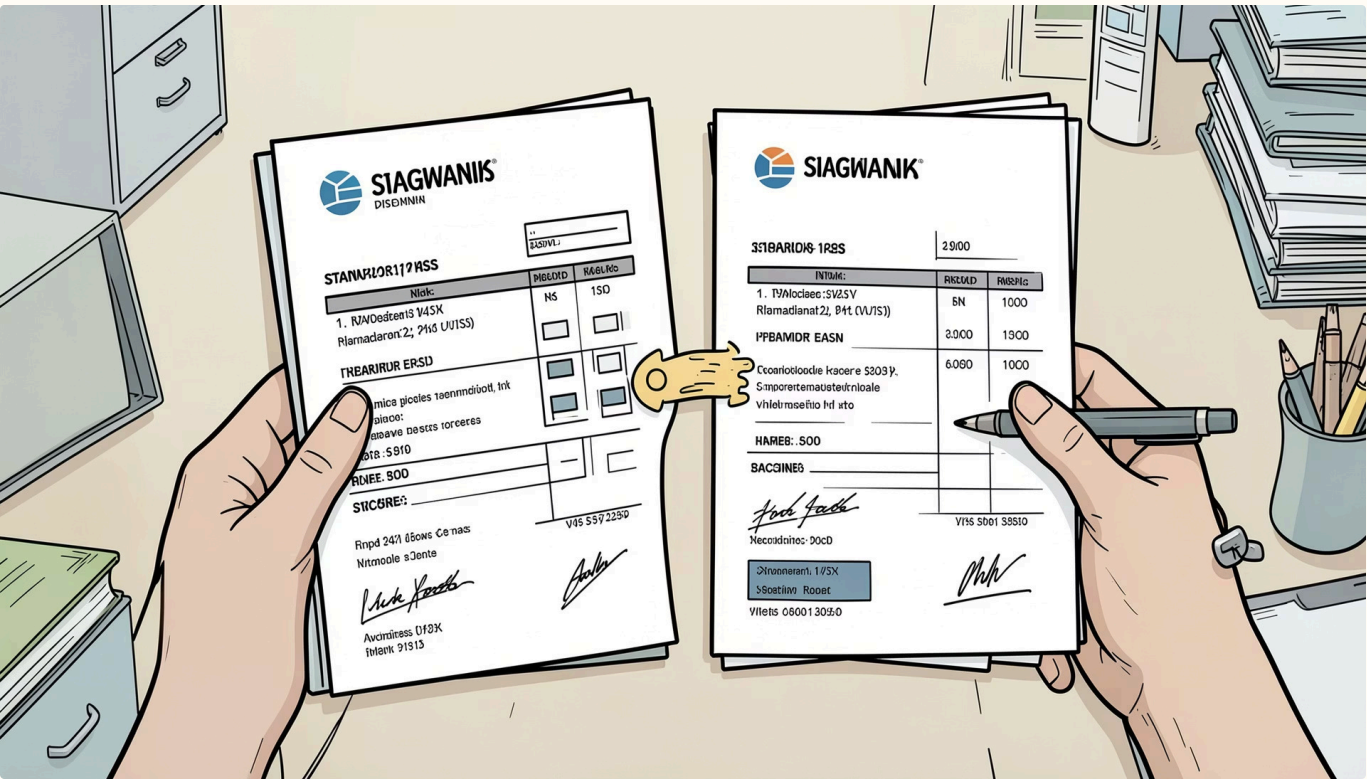
System Administrators

Manage records, oversee data quality, and resolve conflicts.

- ❗ Manual data entries are prioritised over scraped records when conflicts arise, while both versions are retained for full auditability and traceability.

Data Validation and Normalisation Layer

Before data becomes available to users, it passes through a normalisation engine that resolves inconsistencies across providers. Different providers describe the same products using different terminology, making standardisation essential for accurate comparison.



The Problem

Different providers describe products differently. For example:

The normalisation layer standardises these values into a **common comparison structure** so users can make fair, like-for-like comparisons.

Bank A

ZIPIT Transfer Fee

Bank B

Interbank Transfer Charge

Key Normalisation Functions



Data Standardisation

Converts product names, fee structures, categories, and units into common formats for consistent comparison.



Duplicate Detection

Identifies and resolves duplicate records originating from multiple data sources.



Data Matching

Compares scraped records against manual records for the same product to ensure consistency.



Confidence Scoring

Each record receives a confidence score — Manual Upload: **95%**, Scraper: **80%** — to indicate data reliability.



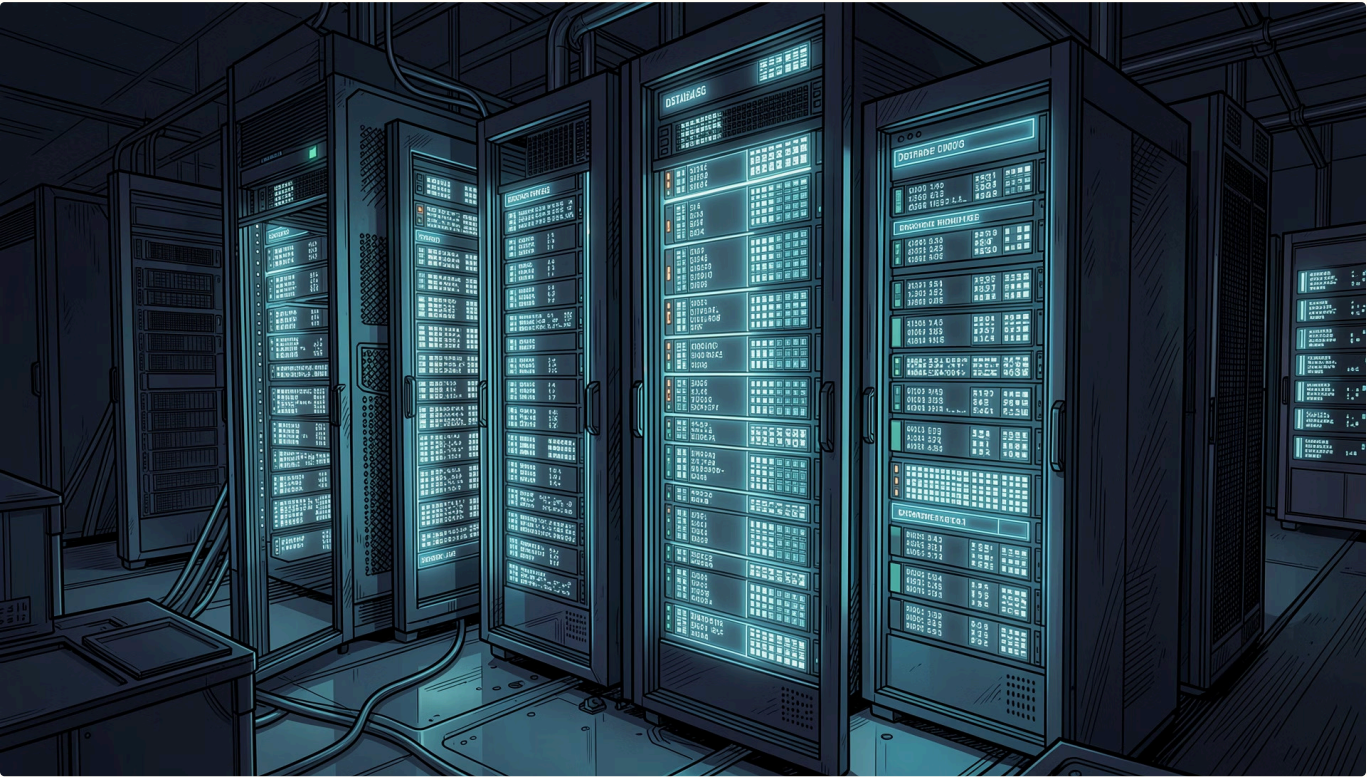
Priority Resolution

When conflicts occur, manual records are prioritised. Both versions are stored for auditability and traceability.

Master Product Database

NEON POSTGRESQL

The Master Product Database acts as the system's **single source of truth**. Built on Neon PostgreSQL, it stores all validated and normalised product data across every supported sector.



Product Data Stored

Financial

Accounts & Loans

Insurance

Policies & Premiums

Telecom

Bundles & Tariffs

Education

Schools & Universities

Hospitality

Hotels & Services

✔ All data stored in the Master Product Database has passed validation and normalisation, ensuring only high-quality records are surfaced to users.

Vector Database & RAQ Architecture

The platform uses a **Retrieval-Augmented Generation (RAG)** architecture to allow AI to retrieve verified product information before generating responses — ensuring accuracy, transparency, and explainability in every AI interaction.



How It Works

01

User Query

User submits a question, e.g. *"Which bank account is best for me?"*

02

Retrieval

Backend sends the query to the retrieval engine, which searches the vector database.

03

Augmentation

Retrieved verified product information is added to the prompt context.

04

Generation

OpenAI GPT generates a grounded response using only verified information.

ML Recommendation Engine & Backend Services

ML Ranking Engine

The platform uses machine learning ranking models to **recommend** products rather than simply display them.

Inputs

- User preferences & budget
- Product category & attributes
- Historical interactions

Outputs

- 1

Best Match
- 2

Best Value
- 3

Lowest Cost
- 4

Highest Rated
- 5

AI Recommended

Backend Services Layer

Provides all business logic powering the platform's core operations.

Product Service Manages all product records.	Search Service Handles search requests.
Comparison Service Generates product comparisons.	Recommendation Service Runs ML ranking algorithms.
AI Chat Service Supports conversational AI.	Analytics Service Generates insights and reports.

Front-End Channels & User Roles

The platform serves multiple distinct audiences through dedicated portals, each tailored to specific needs — from individual consumers to regulators and platform administrators.



Consumer Portal

- Product search & comparisons
- Personalised recommendations
- AI assistant



Corporate Portal

- Market intelligence
- Competitive analysis
- Business comparisons



Regulatory Portal

- Market monitoring
- Transparency dashboards
- Sector analytics



Admin Portal

- Data & user management
- Product approvals
- Platform operations



Consumer



Corporate User



Regulator



Administrator



Manager



Developer